

RIVERCOURSE AGCM, AB, Canada

WMO: 711020

Lat: 53.0189N

Lon: 110.1019W

Elev: 669

StdP: 93.54

Time Zone: -7.00 (NAM)

Period: 07-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-31.8	-29.0	-35.4	0.1	-31.7	-32.4	0.2	-28.9	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.8	28.1	17.5	26.3	16.8	24.7	16.1	19.6	25.5	18.5	24.1	17.3	22.6	N/A	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.4	13.5	23.0	16.3	12.5	21.4	15.1	11.6	20.1	59.0	25.8	55.0	24.1	51.3	22.7	23.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
N/A	N/A	N/A	-35.8	31.0	1.8	1.8	-37.1	32.3	-38.1	33.3	-39.1	34.3	-40.4	35.6		
			-35.9	22.1	1.8	1.3	-37.2	23.0	-38.2	23.7	-39.2	24.5	-40.5	25.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	2.2	-12.6	-13.3	-6.0	2.6	10.8	15.0	17.1	15.8	11.3	3.8	-5.6	-13.4
	DBStd	12.85	8.42	7.79	7.50	5.38	4.54	3.29	2.72	3.13	4.61	4.95	7.00	7.87
	HDD10.0	3562	702	652	497	228	46	4	0	1	38	200	468	727
	HDD18.3	5929	960	886	755	473	234	107	57	89	214	451	718	985
	CDD10.0	718	0	0	0	4	72	155	221	181	77	8	0	0
	CDD18.3	41	0	0	0	0	2	7	19	11	3	0	0	0
	CDH23.3	667	0	0	0	2	65	103	231	172	92	3	0	0
	CDH26.7	127	0	0	0	0	8	16	44	33	26	0	0	0
Wind	WSAvg	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Precipitation	PrecAvg	403	18	12	16	29	44	73	78	57	33	17	15	19
	PrecMax	593	39	28	42	82	114	143	213	169	92	41	52	32
	PrecMin	216	4	2	3	3	5	12	16	5	2	2	2	5
	PrecStd	91	10	6	8	22	29	31	44	30	25	10	12	8
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	4.5	4.1	11.8	21.6	27.7	28.9	30.3	29.6	29.7	22.1	12.0	4.4
		MCWB	1.8	1.8	5.5	9.3	13.5	16.8	19.8	17.9	16.2	12.0	6.2	1.4
	2%	DB	2.9	2.1	7.8	17.8	25.3	26.1	27.7	27.2	25.9	16.9	7.6	2.1
		MCWB	0.9	0.4	3.7	8.1	12.7	15.8	19.5	17.6	14.5	9.3	3.9	0.2
	5%	DB	1.4	0.3	4.9	14.7	22.5	24.1	25.9	25.1	22.8	14.0	4.8	-0.4
		MCWB	-0.3	-1.0	2.3	6.7	11.5	15.1	18.4	17.1	13.6	7.9	2.2	-2.0
	10%	DB	-0.8	-1.6	2.9	11.5	20.1	22.2	24.1	23.2	19.8	11.3	2.6	-2.8
		MCWB	-2.0	-2.8	0.9	5.1	10.6	14.2	17.5	16.4	12.2	6.5	0.7	-3.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	2.1	2.1	6.3	10.2	14.8	18.8	22.1	20.8	17.0	12.7	7.1	1.8
		MCDB	4.2	3.9	10.9	20.3	24.1	25.2	28.0	26.3	26.9	20.9	10.9	3.9
	2%	WB	1.1	0.5	3.9	8.5	13.4	17.3	20.2	19.0	15.3	10.6	4.4	0.2
		MCDB	2.8	1.9	7.4	17.1	22.2	23.9	26.2	24.9	24.4	15.2	7.4	2.2
	5%	WB	-0.3	-0.9	2.3	6.9	12.3	16.1	19.1	17.8	14.0	8.6	2.2	-1.9
		MCDB	1.2	0.3	4.8	13.4	20.7	21.9	24.8	23.4	21.7	13.2	4.2	-0.4
	10%	WB	-2.0	-2.8	1.0	5.5	11.3	15.1	18.0	16.7	12.7	7.0	0.8	-3.9
		MCDB	-0.8	-1.6	2.8	10.9	19.1	20.7	23.0	22.0	19.0	11.3	2.6	-2.8
Mean Daily Temperature Range	5% DB	MDBR	9.2	9.7	9.3	10.9	12.9	11.3	11.8	12.2	12.7	10.6	8.1	8.0
		MCDBR	9.7	10.5	11.0	16.5	16.6	14.6	14.6	16.0	17.9	15.2	10.7	10.4
		MCWBR	8.2	9.1	7.7	8.8	7.3	6.6	7.7	8.0	8.1	8.6	7.4	8.9
	5% WB	MCDBR	9.8	10.3	10.6	15.4	14.4	12.3	13.5	13.8	16.5	14.4	9.7	10.4
		MCWBR	8.4	9.0	7.5	8.5	6.8	6.4	7.7	7.5	7.8	8.4	7.1	9.1
Clear-Sky Solar Irradiance	taub	0.242	0.285	0.284	0.328	0.350	0.366	0.363	0.373	0.311	0.276	0.250	0.221	
	taud	2.338	2.236	2.396	2.371	2.357	2.341	2.380	2.327	2.477	2.548	2.452	2.369	
	Ebn at Noon	773	820	914	907	899	884	880	848	874	846	765	744	
	Edh at Noon	56	86	91	107	115	118	112	111	84	63	50	44	
All-Sky Solar Radiation	RadAvg	1.03	2.09	3.58	4.72	5.81	5.94	6.20	5.04	3.57	2.09	1.02	0.76	
	RadStd	0.10	0.15	0.28	0.43	0.42	0.43	0.34	0.40	0.38	0.30	0.12	0.07	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (53 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page